WORKED SOLUTION

STEP II Specimen paper, Question 9

Pure • Vector geometry

STEP II Specimen paper, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q9

The key idea

Both parts are about one picture: a plane and the perpendicular dropped onto it. In (i), the sphere is centred at the origin, so everything is decided by the foot of the perpendicular from the origin to the plane: its distance from O settles whether the two meet, and Pythagoras then gives the circle. In (ii), split $\mathbf{x}$ into a part along $\mathbf{c}$ and a part perpendicular to $\mathbf{c}$, and watch what the formula does to each.

Solution

Part (i)

The plane $(\mathbf{x}-\mathbf{a})\cdot\mathbf{b}=0$ passes through the point $\mathbf{a}$ and has normal $\mathbf{b}$. Let N be the foot of the perpendicular from the origin O to the plane. Since ON is along the normal, $\mathbf{n}=\lambda\mathbf{b}$ for some scalar λ, and since N lies on the plane,

$$(\lambda\mathbf{b}-\mathbf{a})\cdot\mathbf{b}=0 \quad\Longrightarrow\quad \lambda=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2},$$

which is allowed because $\mathbf{b}\neq\mathbf{0}$. So

$$\mathbf{n}=\frac{(\mathbf{a}\cdot\mathbf{b})\,\mathbf{b}}{|\mathbf{b}|^2}, \qquad d=|\mathbf{n}|=\frac{|\mathbf{a}\cdot\mathbf{b}|}{|\mathbf{b}|},$$

where d is the distance from O to the plane.

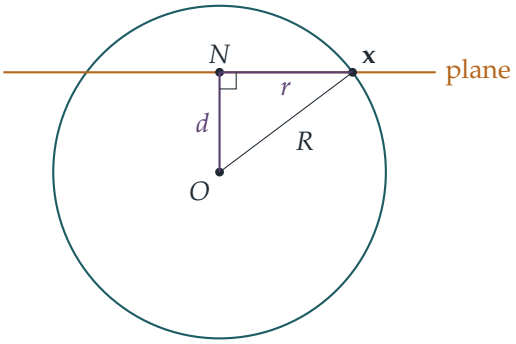
Now take any point $\mathbf{x}$ on the plane. The vector $\mathbf{x}-\mathbf{n}$ lies in the plane, so it is perpendicular to $\mathbf{b}$ and hence to $\mathbf{n}$. By Pythagoras,

$$|\mathbf{x}|^2=|\mathbf{n}|^2+|\mathbf{x}-\mathbf{n}|^2=d^2+|\mathbf{x}-\mathbf{n}|^2. \qquad (*)$$

So points of the plane have $|\mathbf{x}|\geq d$, and every value $|\mathbf{x}|\geq d$ occurs (move away from N within the plane). Hence the sphere $|\mathbf{x}|=R$ meets the plane if and only if $R\geq d$, that is,

$$\boxed{|\mathbf{a}\cdot\mathbf{b}|\leq R\,|\mathbf{b}|} \qquad \text{or equivalently} \qquad (\mathbf{a}\cdot\mathbf{b})^2\leq R^2|\mathbf{b}|^2.$$

In the equality case the plane is a tangent plane and the "circle" shrinks to the single point N.



When the condition holds, the points on both surfaces satisfy $|\mathbf{x}| = R$ and, by $(*)$,

$$|\mathbf{x} - \mathbf{n}|^2 = R^2 - d^2.$$

So they lie in the plane at a fixed distance from N, which is a point of the plane: they form a circle with centre N. Its centre and radius are

$$\mathbf{n} = \frac{(\mathbf{a}\cdot\mathbf{b})\,\mathbf{b}}{|\mathbf{b}|^2}, \qquad r = \sqrt{R^2 - \frac{(\mathbf{a}\cdot\mathbf{b})^2}{|\mathbf{b}|^2}} = \frac{\sqrt{R^2|\mathbf{b}|^2 - (\mathbf{a}\cdot\mathbf{b})^2}}{|\mathbf{b}|}.$$

Part (ii)

Write $\hat{\mathbf{c}} = \mathbf{c}/|\mathbf{c}|$, a unit vector (fine since $\mathbf{c} \neq \mathbf{0}$). Then the relation is

$$\mathbf{x}' = \mathbf{x} - 2(\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}.$$

? **Ask yourself** What is $(\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}$, geometrically?

It is the component of $\mathbf{x}$ along $\mathbf{c}$: the projection of $\mathbf{x}$ onto the line through O in the direction $\mathbf{c}$. So split

$$\mathbf{x} = \mathbf{x}_{\parallel} + \mathbf{x}_{\perp}, \qquad \mathbf{x}_{\parallel} = (\mathbf{x}\cdot\hat{\mathbf{c}})\,\hat{\mathbf{c}}, \qquad \mathbf{x}_{\perp} = \mathbf{x} - \mathbf{x}_{\parallel},$$

where $\mathbf{x}_{\perp}\cdot\mathbf{c} = \mathbf{x}\cdot\mathbf{c} - (\mathbf{x}\cdot\hat{\mathbf{c}})(\hat{\mathbf{c}}\cdot\mathbf{c}) = 0$. The relation becomes

$$\mathbf{x}' = \mathbf{x}_{\perp} - \mathbf{x}_{\parallel}.$$

The component in the plane Π through O perpendicular to $\mathbf{c}$ (the plane $\mathbf{x}\cdot\mathbf{c} = 0$) is unchanged, and the component normal to Π is reversed. That is exactly what a mirror does: **$\mathbf{x}'$ is the reflection of $\mathbf{x}$ in the plane $\mathbf{x}\cdot\mathbf{c} = 0$**, the plane through the origin with normal $\mathbf{c}$.

To confirm this against the definition of a reflection:

- $\mathbf{x}' - \mathbf{x} = -2\mathbf{x}_{\parallel}$ is a multiple of $\mathbf{c}$, so the segment from $\mathbf{x}$ to $\mathbf{x}'$ is perpendicular to Π;
- the midpoint $\frac{1}{2}(\mathbf{x} + \mathbf{x}') = \mathbf{x}_{\perp}$ satisfies $\mathbf{x}_{\perp} \cdot \mathbf{c} = 0$, so it lies on Π.

A few consequences make the picture concrete. Points of Π (where $\mathbf{x} \cdot \mathbf{c} = 0$) are fixed; $\mathbf{c}$ itself goes to $-\mathbf{c}$; $|\mathbf{x}'| = |\mathbf{x}|$, since $|\mathbf{x}'|^2 = |\mathbf{x}_{\perp}|^2 + |\mathbf{x}_{\parallel}|^2$; and applying the map twice gives $\mathbf{x}$ back, as reflecting twice should. The relation is unchanged if $\mathbf{c}$ is replaced by any non-zero multiple $k\mathbf{c}$, which fits: only the plane matters, not the length or sense of its normal.

Checks

In (i), take $\mathbf{a} = \mathbf{b} = (0, 0, 1)$ and $R = 2$: the plane is $z = 1$, which meets the sphere $x^2 + y^2 + z^2 = 4$ in the circle $x^2 + y^2 = 3$, $z = 1$. The formulae give centre $(0, 0, 1)$ and radius $\sqrt{4 - 1} = \sqrt{3}$, as they should. In (ii), take $\mathbf{c} = (0, 0, 1)$: the formula sends (x, y, z) to $(x, y, -z)$, the reflection in $z = 0$. We also tested both parts numerically on 20,000 random choices of $\mathbf{a}$, $\mathbf{b}$, R and $\mathbf{x}$, $\mathbf{c}$: points built on the predicted circle lay on both the sphere and the plane to within 10^{-9}, and the map in (ii) fixed vectors in Π, preserved lengths and dot products, and was its own inverse.

Common slips

- Forgetting the modulus: the condition is $|\mathbf{a} \cdot \mathbf{b}| \leq R|\mathbf{b}|$, not $\mathbf{a} \cdot \mathbf{b} \leq R|\mathbf{b}|$; a plane on the far side of O has $\mathbf{a} \cdot \mathbf{b} < 0$.
- Giving $\mathbf{a}$ as the centre of the circle. The point $\mathbf{a}$ is just some point of the plane; the centre is the foot of the perpendicular from O.
- In (ii), saying "reflection in $\mathbf{c}$" or "in the line along $\mathbf{c}$". Reflection in that line would be $\mathbf{x} \mapsto 2\mathbf{x}_{\parallel} - \mathbf{x}$, which is $-\mathbf{x}'$.

Beyond STEP

The map in (ii) is a *Householder reflection*, $\mathbf{x} \mapsto (I - 2\hat{\mathbf{c}}\hat{\mathbf{c}}^{\mathsf{T}})\mathbf{x}$, a linear map that is orthogonal and symmetric, with determinant -1. Every rotation of space is a product of two such reflections, and numerical analysts use them to factorise matrices stably (the QR decomposition). Strang's *Introduction to Linear Algebra* covers projections, reflections and orthogonal matrices from exactly this point of view.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.